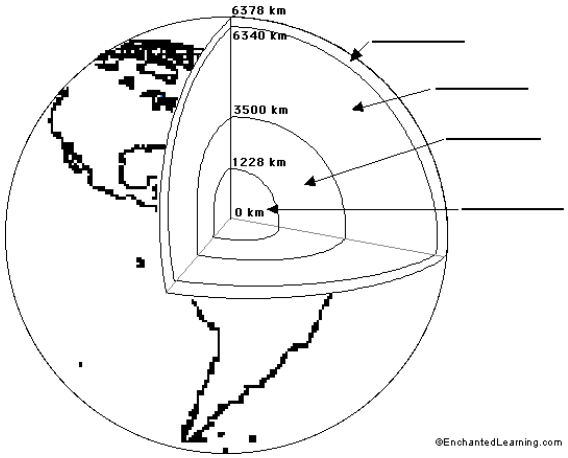
Year 8 Science EXAM REVISION – Term 2 – Rocks

Complete the following questions in your notebooks, using your notes and textbooks

1. Label the diagram of the Earth's layers below:



2. Explain the difference between 'minerals' and 'rocks'.

3. Minerals have six main properties and all are contained in the list of twelve properties below. Identify the six properties which are used to identify minerals.

a) Brightness
b) Lustre
c) Streak
d) Rigidity
e) Cleavage
f) Fracture
g) Clarity
h) Hardness
i) Density
j) Texture
k) Odour
l) Colour

4. Explain the difference between magma and lava.

5. Identify which type of rock (igneous, sedimentary or metamorphic) is being described by each statement below:

a) has layers _____
b) has pebbles all cemented together _____
c) has distinct interlocking crystals _____
d) formed by cooling molten magma _____
e) formed by changing existing rocks with heat and pressure _____
f) may contain fossils _____

6. Explain the relationship between cooling rate of molten rock and crystal size.

7. Explain how you can tell the difference between an intrusive and an extrusive igneous rock>

8. **a)** What is a fossil?

b) How are fossils formed?

9. Complete the table below to summarise the three types of rocks:

TYPE OF ROCK	HOW ARE THEY FORMED?	What makes this rock look different from the other two types of rocks
Igneous		
Sedimentary		
Metamorphic		

10. Explain which type(s) of rock you would expect to find fossils in, and which type(s) you would not expect to find fossils in.

11. a) Define the term *ore*

12. Classify the following rocks as either Igneous, Sedimentary or Metamorphic:

a) Shale
b) Granite
c) Mudstone
d) Marble
e) Basalt
f) Conglomerate
g) Pumice
h) Slate

13. There are six substances or objects mentioned in the list below. Use the information in the list to put the six substances mentioned (*in italics*) in order from hardest to softest:

- A *steel file* can scratch a *pen knife blade*
- *Calcite* scratches a *copper coin*
- A pen knife blade scratches calcite
- *Quartz* can leave a mark on a steel file
- Quartz is scratched by *topaz*

14. An opal miner has been keeping records of the depth of her mine and the total number of opals which have been found in the mine. The data is shown in the table below:

Mine depth (m)	number of opals found
15	15
20	40
25	75
30	90
35	120
40	130
45	135
50	139

a) Use the information in the table below to plot a line graph:

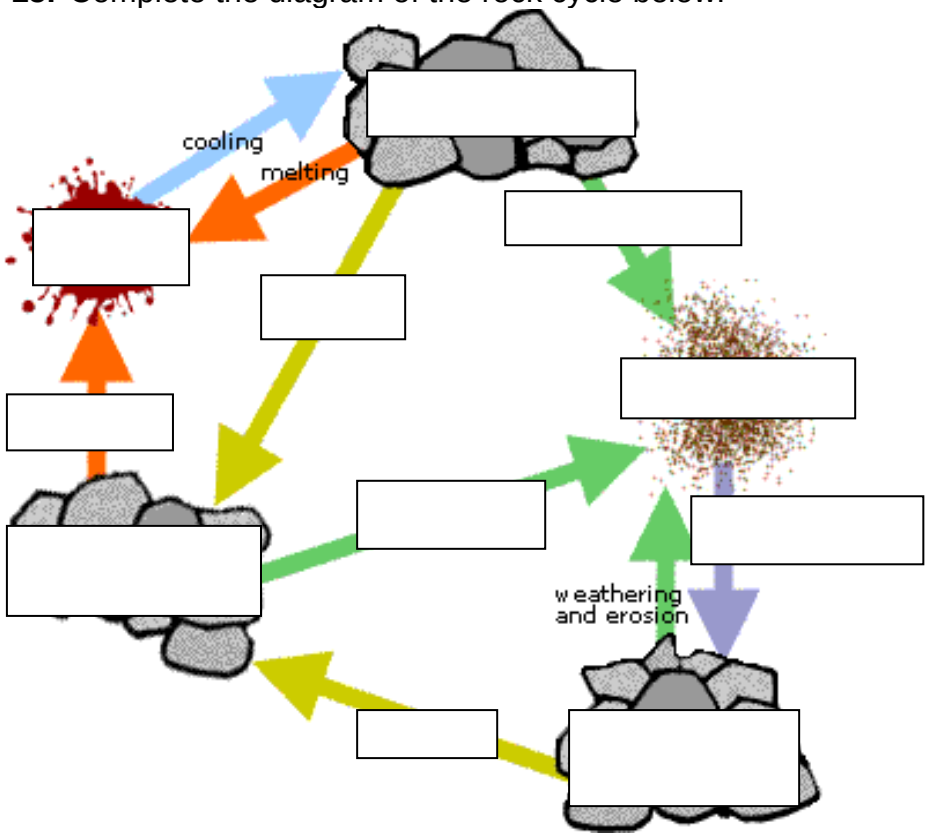


b) State the independent variable.

c) State the dependent variable.

d) Write a conclusion about the information shown in the graph. Be as detailed as possible.

15. Complete the diagram of the rock cycle below:



16. Identify which type of rocks (igneous, sedimentary, or metamorphic) would take the longest to form, and explain why.